Journal of Economic Behavior and Organization 238 (2025) 107202

Contents lists available at ScienceDirect

Journal of Economic Behavior and Organization

journal homepage: www.elsevier.com/locate/jebo

Cannabis deregulation and policing☆

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ARTICLE INFO

JEL classification:
K42

Keywords:
Cannabis
Decriminalization
Racial disparities
Drug use
Drug legalization
Urban

ABSTRACT

Drug crimes continue to make up a large share of the offenses for which individuals interact with the criminal justice system in the United States, with Black Americans arrested at four times the rate of white Americans despite similar drug usage rates. In recent years, policymakers in jurisdictions across the country have deregulated recreational cannabis use, often with the explicit intention of reducing drug crime arrest disparities. Yet, causal evidence about the impact of deregulation on who police arrest is limited. In this paper, we exploit the rollout of the most widespread deregulatory approach related to recreational cannabis use—the decriminalization of cannabis possession—across the three largest US cities, New York City, Los Angeles, and Chicago, using a difference-in-differences design. We find that decriminalization significantly reduced cannabis possession arrests. We observe that decriminalization narrowed racial disparities in arrests in Chicago by reducing small quantity possession arrests for Black individuals and in Los Angeles by reducing large quantity possession arrests for both Black and Hispanic residents. Lastly, we extend our analysis to legalization of recreational cannabis use and observe that legalization decreased arrests for every racial and ethnic group we consider, with similarly large impacts across groups.

Until recently, over 500,000 Americans were arrested annually for possession of cannabis, making up 5% or more of all arrests across the country (Neill Harris, 2022; Ingraham, 2017). This landscape is rapidly changing, as cannabis possession has become one of the fastest evolving major state-level legislative areas of the past century (Tribou and Collins, 2015). Meanwhile, racial disparities have persisted even amidst the increasing legal leniency, where Black individuals are four times as likely to be arrested as white individuals though usage rates are similar across the two groups (Substance Abuse and Mental Health Services Administration, 2022; American Civil Liberties Union, 2020). With an arrest potentially leading to detrimental subsequent outcomes for individuals in a range of key aspects, a fundamental question is how the changing judicial landscape impacts different racial and ethnic groups' engagement with the criminal justice system.

In this paper, we use data from the three largest American cities (New York City, Los Angeles, and Chicago) to estimate the impact of the country's most widespread recreational cannabis policy, the decriminalization of the small quantity possession of cannabis. In all three geographical areas we examine, during the 2010s, decriminalization took place, meaning that the small quantity possession of cannabis for recreational purposes ceased to be an arrestable offense, and instead became a violation punishable with a fine only. Using a difference-in-differences framework, we examine the policy rollout in each city, exploiting the fact that regulatory changes about cannabis possession were independent of and often enacted at different times than regulation of other crimes, including other drug crimes.

☆ We thank Erin Wright for excellent research assistance. We thank Ashna Arora, Tyler Giles, Seth Neller, Christian Saenz, and conference participants at the American Society of Health Economics (ASHEcon) annual conference for exceedingly helpful comments. We thank the Society for Economics Representation and Networking for a workshop participation stipend related to this project.

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https://doi.org/10.1016/j.jebo.2025.107202
Received 10 January 2024; Received in revised form 6 August 2025; Accepted 9 August 2025
Available online 16 September 2025

We find that, in all three cities, decriminalization led to a significant—though incomplete—reduction in arrests for deregulated offenses. In addition, we observe that decriminalization improved racial disparities in arrests in Chicago by reducing small quantity possession arrests for Black individuals, and in Los Angeles by reducing large quantity possession arrests for both Black and Hispanic residents. Lastly, we estimate the impact of subsequent legalization in New York City and find that it decreased arrests for the deregulated offense both overall and for every racial and ethnic group we study.

An important driver for deregulation in each city was the disproportionate impact cannabis prosecution has on Black Americans and other communities of color (Mendez, 2021; City of New York, 2023; State of Illinois, 2019; Matthews, 2014). Cannabis decriminalization was often framed as a step in ensuring that the criminal justice system reaches *fewer* Black Americans and other communities of color in a *fairer* way. Given the findings we observe, we conclude that decriminalization has the potential to be an effective, though partial step toward achieving the above goals identified by policymakers, but only legalization realizes that potential fully. One context to consider this finding in is that nearly half of Black Americans who interacted with the police recently viewed that interaction as not a positive experience (Lloyd, 2020), and that the lifetime risk of arrest for crimes broadly is twice as high for Black versus white Americans (Barnes et al., 2015).

Our study advances the literature in multiple ways. First, our work ties to the growing literature on alternatives to arrest, charging, and prosecution for a range of crimes, where studies find that reducing individuals' engagement with the criminal justice system has a neutral to positive impact on subsequent rearrests (Agan et al., 2023; Mueller-Smith and Schnepel, 2021; Drago and Galbiati, 2012; Arora and Bencsik, 2021). We expand this literature by assessing the impact of decriminalization policies and by estimating the causal impact of cannabis-related deregulatory steps on arrests across multiple cities.

Second, this study contributes evidence to the literature on racial disparities in the criminal justice system (Feigenberg and Miller, 2020; Lofstrom and Raphael, 2016; Cox and Cunningham, 2021; O'Flaherty and Sethi, 2019; Agan, 2022) by addressing drug arrests, an important component among offenses leading Black Americans and other communities of color to interact with the criminal justice system. Further, we focus on arrests, the first step of a criminal justice system interaction, where for example Rehavi and Starr (2014) find that these initial circumstances (the arresting offense and the individual's criminal history in particular) explain the vast majority of racial disparities in time spent in prison by Black versus white individuals. The current study's evidence that the decriminalization of cannabis possession is able to partially reduce Black Americans' interactions with the criminal justice system can have substantial implications for all contexts beyond arrests, such as courts and the correctional system.

Third, we contribute to the literature by examining the effectiveness of one of the most often used policy levers across the legislative branch. Increasing or decreasing penalties is one of the most frequently used levers in settings ranging from environmental protection through tax return preparer behavior to the payment of traffic fines (Blondiau et al., 2015; Welisch, 2018; du Plessis et al., 2020; Anderson and Cuccia, 2000).

Lastly, engagement with the criminal justice system has detrimental effects on individuals' subsequent outcomes. Much of the assessment of this focuses on the impact of convictions, finding that incarceration reduces employment odds and wages (Mueller-Smith, 2015; Finlay, 2008), especially for Black job applicants (Pager, 2003), though the length of incarceration does not meaningfully affect these outcomes (Kling, 2006). But even arrests themselves lead to adverse outcomes, with for example the income of arrestees decreasing in the quarter of arrest (Grogger, 1995). Given these patterns, this study's findings can have important implications for crime policy and policies intending to address racial disparities.

The remainder of this paper is divided into five sections. Section 1 describes the policy evolution of decriminalization, Section 2 details the data sources used and the descriptive patterns we observe regarding deregulated cannabis offenses, Section 3 describes the empirical strategy applied, Section 4 presents the findings, and lastly, Section 5 concludes.

1. Setting

Across the country, between 2001 and 2010, over half of all drug arrests were for cannabis, and nine out of ten of those were for possession rather than sale. Black individuals were on average four times as likely to be arrested for possession as white individuals when the two racial groups use the substance at equal rates (Substance Abuse and Mental Health Services Administration, 2022; American Civil Liberties Union, 2020). Usage rates among Hispanic individuals are 20%–25% lower than among Black and white individuals.

Meanwhile, public opinion grew steadily more supportive of cannabis consumption during the past decade. In 2011, nationwide support versus disapproval for legalization was exactly 50%–50% among Americans, then reached a majority in 2013 with 58% of Americans supporting it. 2020 marked the first year when more than two-thirds, 68%, of Americans were pro-legalization (Jones, 2022). Understanding how deregulation changes deregulated cannabis arrests and related arrests then is increasingly relevant to a large share of the US population.

Motivations for deregulation have been multi-fold, such as economic considerations regarding tax revenue, but a growing, central argument present among policymakers has been improving racial equity. The argument that Black, Hispanic, and other individuals of color are over-arrested in proportion to the presence of drug use across racial groups has been a tenet of recent policymaker motivations for decriminalization.[1] For example, on the day of ratifying decriminalization in New York, then-Governor Cuomo stated that "Communities of color have been disproportionately impacted by laws governing cannabis for far too long, and today we are

[1] California and New York state have a Black-to-white cannabis possession arrest ratio of 1.81 and 2.63, respectively, while Illinois' ratio is the third highest in the nation with 7.51 (American Civil Liberties Union, 2020).

ending this injustice once and for all" (Campbell, 2019; Community Service Society, n.d.). Policies in the earlier 2010s were framed more as a mix between fairness and financial scrupulousness, for example with Illinois' Rep. Kelly Cassidy, the state representative introducing the bill that became decriminalization law in the state, noting the importance of "creating cannabis policies that treat our residents more fairly and free law enforcement up for more serious crime" (Garcia, 2016a). Meanwhile in California, which jurisdiction's decreased penalty was the earliest deregulatory lever discussed here, deregulation was framed predominantly as a cost-saving mechanism (Condon, 2010).

1.1. The decriminalization of cannabis

Policy levers for limiting criminal justice system responses to cannabis possession can take two core legal approaches, the decriminalization and the legalization of low-level cannabis offenses. In part because the former requires less extensive policy agreement across parties, it has been the most widespread policy lever in the country for the past decades, with 26 states and Washington, D.C., implementing a decreased penalties (generally, decriminalization) law by early 2024.[2]

Decreased penalties is defined as changing the legal consequences of an act from a more severe outcome to a less severe one. More specifically, decreased penalties when it comes to drug use can take the form of either reclassifying an offense into a criminal or a civil penalty (i.e., fine) that is a violation rather than a crime—with neither penalty format carrying the potential for jail. Most states opted to make the fines a civil penalty, thereby enacting the decriminalization of small quantity possession of cannabis (Hartman, 2023). However, a handful of states kept small quantity possession as a criminal offense but removed the possibility of jail time.[3]

Across California, the decriminalization of cannabis possession occurred in September 2010, when state lawmakers reduced the punishment for small quantity cannabis possession from a misdemeanor to an infraction, with a monetary penalty of up to $100 (McKinley, 2010). Illinois experienced a similar shift when small quantity possession was downgraded to an infraction resulting in a fine of $100–200 starting July 2016 (Garcia, 2016b).[4] Lastly, in 2019, New York lawmakers reduced the penalty for possession of under 2 ounces of cannabis as well as public consumption to a violation resulting in a fine of $50 to 200.[5] Table A1 summarizes the exact weight limits decriminalized and the announcement and implementation dates of decriminalization in each city. Notably, none of the ratification dates fall on January 1st, when many laws tend to commence, and we find that no other major drug laws were ratified in the respective jurisdictions in the same month as the cannabis laws we study, supporting our identifying assumption that, in the absence of the cannabis decriminalization law, cannabis and non-cannabis arrests would have evolved similarly.[6]

Prior European evidence by Adda et al. (2014) on decreased penalties for drug offenses, using a one-year-long pilot policy in one London borough where offenders were given a warning rather than being arrested, found mixed effects, with the number of cannabis possession offenses staying flat in the first half of the year when the policy was framed as a temporary experiment, and increasing in the second half and beyond once an extension of the policy was announced. Meanwhile, possession arrests decreased throughout the observation period.

The decriminalization of cannabis might impact possession offenses above the decriminalized threshold as well, potentially due to including individual as well as police behavioral changes. On the one hand, those possessing cannabis might start to opt to carry below-threshold cannabis weights. Notably, all three jurisdictions decriminalized the possession of cannabis equivalent to at least approximately twenty cannabis cigarettes, or almost a week's worth to several weeks' worth of substance for the regular user (Chokshi, 2016; Bartlett, 2022). Additionally, evidence to date suggests that cannabis use increases after legalization specifically, potentially implying that more cannabis could be carried in general after deregulatory steps (Hollingsworth et al., 2022; Zellers et al., 2022; Midgette and Reuter, 2020). While theoretically either behavior change is possible, to our knowledge, no research evidence exists to date on de facto behavioral changes.

[2] This paper focuses on adult/recreational use cannabis laws and does not address medical cannabis legislation's impact on arrests for three reasons. First, a very small share of the population had or has medical cannabis licenses. At the time when adult use was not (yet) legal, across states this share ranged from below 0.01% (in Minnesota and Delaware) to 2.3% (in Oregon); and three-quarters of medical cannabis users used cannabis (illegally) before getting a medical cannabis card. Second, white individuals are vastly overrepresented among medical cannabis users, potentially suggesting that it is adult use rather than medical use policies that are of relevance from the policymaker perspective regarding decreasing racial inequities (Fairman, 2016; Rosenthal and Pipitone, 2020; Hartman, 2023). Third, as Dave et al. (2025) discuss, medical and recreational cannabis laws target different populations and might impact outcomes differently. We direct the reader to Anderson and Rees (2023) for a recent review on the state-specific processes of medical cannabis legalization and its public health impacts, and to Chu (2014) and Chu and Townsend (2019) regarding the effect of medical cannabis legalization on arrests.

[3] Given that in the vast majority of cases decreased penalties equates decriminalization, we use the two terms interchangeably in this study.

[4] The Chicago City Council in an ordinance lowered penalties for small quantity cannabis possession of up to 15 grams in 2012, four years before the state did. However, arrests continued to make up the vast majority of criminal justice responses to small quantity possession (over 92% of instances involved arrests, while less than 8% of instances involved tickets) and continued to impact predominantly people of color (in 96% of arrests), suggesting that the policy wasn't meaningfully implemented in practice (Dumke, 2014; Mack, 2012). As Chicago's accessible arrest data only goes back to 2014, we are unable to further quantify the city ordinance's impact in this study.

[5] Previously, possession of over 25 grams (0.89oz) and public consumption were both classified as misdemeanors (Campbell, 2019). Over a year prior to decriminalization, district attorneys in two of the five New York City boroughs initiated nonprosecution, meaning they announced they do not intend to prosecute small quantity cannabis charges in their specific boroughs, followed by the New York Police Department announcing a non-arrest approach for such offenses. Regardless, the offense remained a criminal act and substantial numbers of arrests continued (on average, 142 per month in the 6 months prior to decriminalization), making the law change particularly interesting to examine.

[6] Illinois amended its cannabis Driving Under the Influence (DUI) law at the same time as decriminalizing small quantity possession. The amendment moved from considering any amount of cannabis to qualify for a DUI to setting a specific level of blood tetrahydrocannabinol (THC) to qualify for the offense; however, at the time, there was no lab across the state that could measure THC in the way the law set out to measure it (Ivec, 2016).

On the other hand, it is possible that police officers' approaches or crime-detecting abilities shift after deregulatory steps, which is supported by existing evidence with regards to legalization specifically. Iannacchione and Ward (2022) in the context of Colorado find that after legalization, marijuana offenses were not a priority for police departments—in fact, 51% of police respondents strongly disagreed with the statement that "marijuana arrests for possession of any amount of marijuana are a priority for [their] department." Similarly, in the context of Oregon, years after the legalization policy commenced, 68% of officers agreed that determining when an individual possesses an illegal amount of marijuana is very difficult or difficult (Henning et al., 2023). Thus, it is possible that spillovers in arrests may be in part related to both police department attitudes and enforcement abilities.

2. Data

Data comes from each city's Data Portal. Data for each city contains the date of the arrest, the race and ethnicity of the arrestee, and the offense associated with the arrest, both as a locality-specific offense code and as a short text description. Using these data, we construct the monthly count of arrests for (a) decriminalized cannabis-related offenses (small quantity possession, as well as in the case of New York City, consumption in public view), (b) the two types of cannabis-related offenses that are not deregulated (large quantity possession and sale), and (c) other crime categories.

We identify decriminalized cannabis offenses in each city by identifying the location-specific offenses that are below the decriminalized weight limit. In New York City, a range of different offense types (such as possession from fifth to first degree, among others) appear in the data, which can be clearly mapped to the New York Penal Law and the New York Codes, Rules and Regulations (New York State Penal Law, n.d., 2021; New York State Register, 2019). From these, we precisely identify small quantity possession offenses, as well as the New York specific charge, the consumption of cannabis in public view, that were decriminalized.[7] In Los Angeles, similar to New York City, we are able to match offense codes specifically to the decriminalized offenses. We do observe one cannabis offense code in the Los Angeles data—making up 10% of all cannabis possession arrests—that does not provide any information on weight of the drug, meaning it is not possible to know whether this offense code is deregulated or not. Because over two thirds of cannabis possession arrests in the city are for weight categorizes not subsequently deregulated, we conservatively opt to classify this offense as non-deregulated as well. Lastly, in Chicago, we are again able to match the specific offenses to the decriminalized weight limit of 10 grams. We do have one offense code that is possession up to 15 grams, which we include as a treated category because two thirds of the weight range it covers is treated—arrests for this code make up a small portion, 8% of treated arrests.[8]

As for non-deregulated cannabis offenses, in all three cities we identify large quantity possession (above the city-specific decriminalization limit) and the sale of cannabis offenses as offenses that remained illegal once deregulation took place.[9] Finally, we construct control categories as follows. Because each city has slightly different offense categories based on local laws, in order to largely harmonize the type and number of control groups, we apply the Federal Bureau of Investigation's Uniform Crime Reporting (UCR) Handbook (Federal Bureau of Investigation, 2004). Then, we split each UCR crime type into felonies and misdemeanors (where the latter also includes any potential other local non-felony categories), resulting in a total of at least 36 control categories. In Los Angeles, the available dataset contains a crime category variable corresponding to the UCR codes. In New York City, we hand-match offenses to the broader UCR codes.[10] In Chicago, we supplement the public arrest data with one additional variable from a more detailed arrest dataset (also available through the Data Portal), which contains the FBI UCR code for each offense (this additional variable is less detailed than the main dataset's offense information, but removes the necessity to hand-match local codes to the broader UCR categories).

2.1. Descriptive statistics

Descriptive statistics regarding cannabis and control arrests in the three cities are presented in Table 1. Capturing the 6 months prior to decriminalization, we observe that there were on average 18,033 arrests across New York City. 142 of those were for to-be-decriminalized cannabis offenses, 4% of which are received by white arrestees and 96% of which are received by arrestees of color.[11] In Los Angeles, 11,798 arrests occurred on average, of which 77 were for to-be-decriminalized cannabis offenses. 21% of

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[7] Prior to decriminalization, New York City had reduced the penalty for some cannabis possession offenses of up to 25 grams to an infraction in 1977 (Breasted, 1977). However, as some such arrests continued, and because city officials discussed charges involving possession of less than 25 grams as part of the deregulatory efforts in 2019 (The Brooklyn District Attorney's Office, 2018), we consider all arrests up to the deregulated limit of 57 grams treated in the analysis. In practice, only 10% of to-be-decriminalized arrests involved possession of below 25 grams in the year prior to decriminalization.

[8] Data in Chicago lists potentially multiple charges associated with one arrest. To be able to allow results to be consistent across the three cities, we use the first charge information in Chicago, similarly to what is available in the New York City and Los Angeles datasets.

[9] There were a few additional cannabis-related arrest offense categories that rarely occurred in the data and do not meaningfully fall under either large quantity possession or sale of cannabis, such as cannabis offenses related to driving and cultivation of cannabis plants. In total, these rare offenses constitute less than 5% of cannabis arrests, and we exclude these from the analysis.

[10] The universe of UCR codes is not exactly identical to the universe of offense codes in local laws. In Los Angeles, the UCR codes in the data differ from the federal categories in very minor ways, such as fraud and embezzlement being separate UCR codes federally but being grouped into one category in the Los Angeles UCR codes. In New York City, we create matches based on the information available, which is facilitated by local codes being much more detailed than the broad UCR categories. Where offense categories not associated with a specific UCR crime type appear in local data, which is rare, such observations are placed in non-UCR numbered control groups.

[11] We focus on Black, Hispanic, and white individuals here and throughout the study, because they make up over 95% to over 99% of cannabis-related arrests across the three cities.

Table 1
Average monthly arrests prior to decriminalization.

	New York City			
	All	Decriminalized Cannabis Offenses	Untreated Cannabis Offenses	Non-Cannabis Offenses
All Arrestees	18 033	142	181	17 710
Black Arrestees	8458	88	108	8262
Hispanic Arrestees	6140	45	57	6039
White Arrestees	2198	6	6	2186
	Los Angeles			
	All	Decriminalized Cannabis Offenses	Untreated Cannabis Offenses	Non-Cannabis Offenses
All Arrestees	11 798	77	444	11 277
Black Arrestees	3512	22	173	3317
Hispanic Arrestees	5408	35	178	5196
White Arrestees	2258	16	72	2170
	Chicago			
	All	Decriminalized Cannabis Offenses	Untreated Cannabis Offenses	Non-Cannabis Offenses
All Arrestees	5929	21	96	5812
Black Arrestees	4218	18	83	4118
Hispanic Arrestees	1120	3	11	1106
White Arrestees	543	1	2	540

Note: The above table captures the average monthly counts of to-be-decriminalized cannabis offenses, untreated cannabis offenses, and all non-cannabis offenses during the 6 months prior to the city-specific decriminalization. Averages are rounded to the nearest whole number. Due to low counts, arrests for racial and ethnic groups outside of Black, Hispanic, and white individuals are not displayed separately.

these involved white arrestees, while 79% involved Black or Hispanic arrestees, or other racial minorities. Meanwhile, in Chicago, an average of 5929 arrests occurred per month, of which 21 constituted a to-be-decriminalized cannabis offense. 5% involved white arrestees and 95% involved arrestees of color.

While we observe 74%–99% of cannabis-related arrests in these three cities being of Black and Hispanic individuals, these racial and ethnic groups compose 48%–58% of the populations in these cities, with the largest share in Chicago and the smallest share in New York City. Looking at arrests of Black individuals specifically, we see that 29%–86% of cannabis arrests are of Black individuals, with the highest share in Chicago and the lowest share in Los Angeles. Demographically, Black individuals make up 8%–28% of the population in these cities, again with the highest share in Chicago and the lowest in Los Angeles. Considering only Hispanic arrestees, we observe that 14%–45% of cannabis arrests are of Hispanic individuals, with the highest share in Los Angeles and the lowest in Chicago. Population-wise, Hispanic individuals make up 28%–47% of the residents of these cities, with the highest share in Los Angeles and the lowest share in New York City (Frey, 2021).

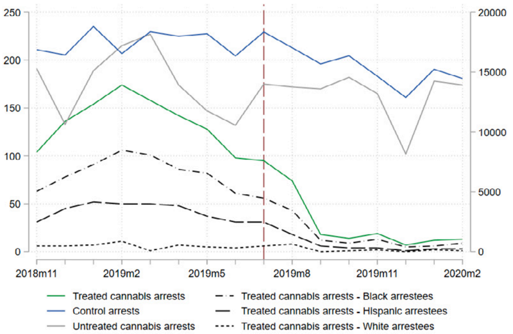
We note that observations in our treatment and control groups are unbalanced, i.e., there are substantially more arrests for all offenses outside of decriminalized cannabis offenses than for decriminalized cannabis offenses. Unbalanced treatment versus control group sizes can reduce the precision of the estimate; however, this is still a preferred approach compared to having balanced and small treatment and control groups. For example, Hutchins et al. (2015) in simulations show that regardless of the extent of balance, the point estimates are stable; however, precision is improved with larger and more balanced treatment and control groups. Such unbalanced groups are often a feature of studies on the criminal justice system that investigate crimes outside of the most frequent crime types.

Descriptively, we summarize the trend over time in the months prior to and following decriminalization in Fig. 1. We capture treated (decriminalized) offenses, including broken out for Black, Hispanic, and white arrestees, as well as non-deregulated cannabis offenses, and control offenses. In all three cities, we observe that decriminalized, control, and non-deregulated offense counts evolve similarly prior to deregulation, and that after decriminalization, the impacted offenses decrease overall, as well as within each racial and ethnic group, while control offenses remain broadly flat and not deregulated offenses somewhat decline.

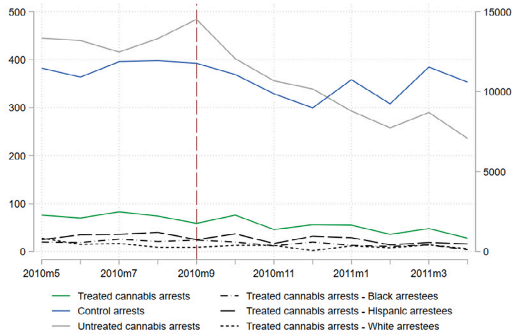
3. Empirical strategy

This section describes the difference-in-differences (DID) framework applied in the analysis to identify the impact of decriminalization on subsequent arrests for cannabis offenses. We compare cannabis-related arrests that the decriminalization targeted to other types of arrests unrelated to cannabis offenses before and after the cannabis-specific deregulatory policy goes into effect.

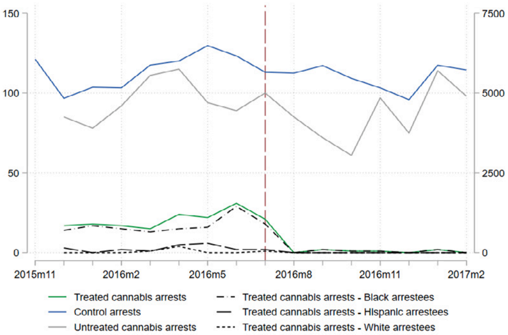
The DID estimate's identifying assumption is that, in the absence of deregulation, cannabis and non-cannabis arrests would have evolved similarly over the observed period. Using all non-cannabis arrests among control offenses is well fitted in this setup, because prior evidence, while somewhat mixed, predominantly suggests that cannabis legalization does not substantially change



(a) Panel A: New York City



(b) Panel B: Los Angeles



(c) Panel C: Chicago

Fig. 1. Cannabis and non-cannabis arrests over time.
Note: Panels A, B, and C plot the number of arrests for decriminalized cannabis offenses, non-decriminalized cannabis offenses, and non-cannabis offenses in New York City, Los Angeles, and Chicago, respectively, in the months prior to and following decriminalization. Decriminalized offense counts and never deregulated offense counts are reflected on the left y-axis, while control (non-cannabis) offense counts are reflected on the right y-axis due to their comparatively larger volume. The vertical red line denotes the first treated month.

the frequency of other crimes, and if it does, evidence is mixed on what type of crime would change.[12] We estimate the impact of the policy lever in each city separately, rather than in a staggered rollout design, such as with a stacked difference-in-differences approach (Baker et al., 2022), to better assess potential differences across jurisdictions.

Central specification

The start of decriminalization is used to estimate the following equation:

$$Y_{c,ym} = \beta_1 Treat_c + \beta_2 After_{ym} + \beta_3 Treat_c * After_{ym} + \varepsilon_{c,ym}$$

$Y_{c,ym}$ is the number of arrests for crime type c in year-month ym. β_1 captures the decriminalized outcomes of interest. $Treat_c$ is an indicator variable that equals one for deregulated cannabis offenses and zero for all other, non-cannabis-related crime types. Because we might expect that the two cannabis-related crimes that were never deregulated across the three cities, large quantity possession and sale, could also experience changes in arrests after decriminalization, we exclude arrests for these offenses from the main analysis. Instead, we address outcomes for these two crimes directly in Section 4.

$After_{ym}$ is an indicator variable that equals one if deregulation has taken place in period ym. Standard errors are clustered at the offense category (UCR combined with felony/misdemeanor) level. In two of the three cities, the decriminalization laws were ratified (signed into law by the governor) 1–3 months prior to going into effect, while in the third city effectiveness was instantaneous with ratification. To account for potential anticipation effects, we consider the ratification date as our treatment date if it occurred before the effectiveness date. We largely focus on the 8 months before and after the policy implementation, where a harmonized 8

[12] Dills et al. (2017), using survey evidence, find no changes in crime, Brinkman and Mok-Lamme (2019) find no changes outside of census tracts with dispensaries, a very much localized effect where crimes decrease, predominantly driven by low-level non-violent crimes, while Dragone et al. (2019) find that crimes do decrease, but the changes are driven by rapes and property crimes. Meanwhile, Hansen et al. (2020) examining specifically traffic fatalities involving cannabis and alcohol find that these remained unchanged.

Table 2
Cannabis decriminalization's effect on arrests for decriminalized offenses.

	New York	Los Angeles	Chicago
Panel A: All Arrestees			
	−53.44**	−3.43	−12.84***
	(21.27)	(7.14)	(3.93)
Pre-Treatment Mean	136.75	75.75	20.62
Percent Change	−39.08	−4.53	−62.27
F-test	0.62	0.01	0.86
N	592	600	752
Panel B: Black, Hispanic, and White Arrestees			
Black Arrestees	−12.57	15.07**	−9.59**
	(21.27)	(7.14)	(3.93)
Pre-Treatment Mean	83.50	21.50	17.00
Percent Change	−15.05	70.08	−56.43
F-test	0.63	0.02	0.60
N	592	600	752
Hispanic Arrestees	17.56	11.44	1.78
	(21.27)	(7.14)	(3.93)
Pre-Treatment Mean	43.00	34.00	2.88
Percent Change	40.83	33.65	62.00
F-test	0.27	0.60	0.03
N	592	600	752
White Arrestees	48.43**	14.19*	3.91
	(21.27)	(7.14)	(3.93)
Pre-Treatment Mean	5.88	17.25	0.62
Percent Change	824.41	82.27	625.22
F-test	0.09	0.00	0.03
N	592	600	752

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.
Notes: Panel A captures changes in the number of arrests for the decriminalized cannabis offenses compared to all non-cannabis offenses. Panel B is identical to Panel A, except treated cannabis arrests are only included if they are of an individual from the specific racial and ethnic group being examined. Control offenses are unchanged.

month post-period in all cities enables us to end the analysis in New York City in February 2020, thus excluding the period of the COVID-19 pandemic (Los Angeles and Chicago implemented the policy in earlier years). (Further, following Miller (2023), in Los Angeles specifically, we use a somewhat shorter pre-period window.) We define the start of treatment as the month during which the policy was ratified, that is, if ratification took place for example on March 15, we consider March the first treated month.

Event study specification

In order to examine the month-by-month impact of deregulation, we use the following event study specification:

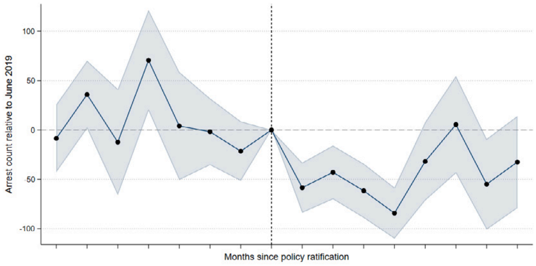
$$Y_{c,ym} = \sum_{\tau \geq -n} \beta_\tau Treat_c * After^{\tau}_{ym} + \gamma_{ym} + \gamma_c + \varepsilon_{c,ym}$$

$Y_{c,ym}$ is defined exactly as above. $After^{\tau}_{ym}$ are indicator variables that equal one if the deregulation occurred exactly τ months before period *ym*. For example, in New York City decriminalization started in July 2019, so $After^1$ indicator equals one for cannabis arrests for July 2019, the $After^2$ indicator equals one for cannabis arrests for August 2019, and so on. γ_{ym} captures a year-month fixed effect, while γ_c captures whether the offense was deregulated or not. Observed values for τ during the pre-treatment period test for any potential differences between patterns for cannabis and other arrests before the policy started.

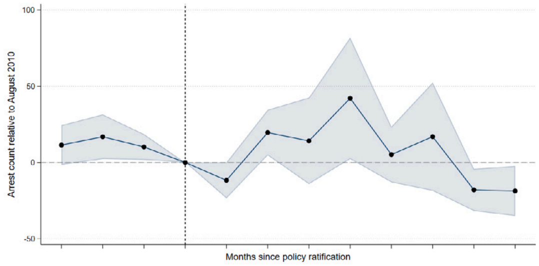
4. Results

Table 2 reports estimates of the impact of decriminalization on arrests for decriminalized offenses in each city. First focusing on all arrestees, we observe that decriminalization significantly decreased the number of arrests for decriminalized offenses in New York and Chicago, with decreases ranging from 39 to 62%. In Los Angeles, we do not observe a significant change in arrests overall, though we note that the pre-trend F-test is significant, so we limit drawing definitive conclusions with regards to overall patterns in Los Angeles.

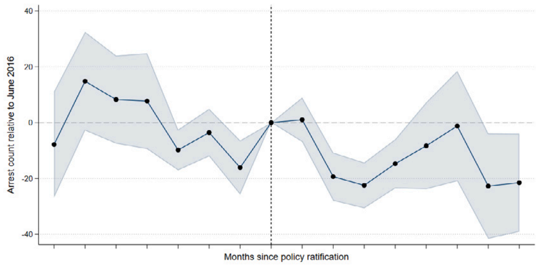
Considering the effects of decriminalization on each racial and ethnic group, we find that arrests for Black individuals decreased in Chicago and remained unchanged in New York. Meanwhile, arrests for Hispanic individuals remain unchanged across the study settings. Arrests for white individuals are lower than for either other racial and ethnic group we examine, where we can cautiously



(a) Panel A: New York City



(b) Panel B: Los Angeles



(c) Panel C: Chicago

Fig. 2. Cannabis deregulation's effect on arrests for deregulated offenses over time.
Note: Panels A, B, and C plot the difference-in-differences for New York City, Los Angeles, and Chicago, respectively. The comparison period in all cities is the month before policy ratification.

speak to patterns post-policy, we find that arrests potentially increase in New York. In Chicago, Black individuals make up 86% of the pre-treatment arrest counts while making up 20% of the city's population. The observed change suggests that decriminalization in Chicago narrowed racial disparities in arrests by reducing arrests for Black Americans. As mentioned in Footnote 1, important context here is that Chicago had more disproportionate arrest rates of Black versus white individuals than the vast majority of the country, with Illinois making 7.5 cannabis arrests of Black individuals for every arrest of a white individual, while this ratio was 1.8–2.6 in New York and Los Angeles.

Fig. 2 shows the temporal patterns in how decriminalization impacted arrests. In both Panels A and C, we observe an immediate, significant decrease in the number of arrests for offenses that decriminalization targeted. In both cities, we observe this decline for approximately 4 months after the policy ratification, then see a slight reversion to the pre-period patterns, though the overall effect remains a decrease, and a decrease continues especially in Chicago. In Los Angeles, we do not observe a meaningful change, which aligns with the regression results.

To test the robustness of our findings, we reproduce our main regression while excluding first the most severe UCR code's felony offenses (UCR code 1 "Criminal Homicide", where all UCR code 1 offenses are felonies due to their severity) and next the least severe UCR code's misdemeanor offenses (UCR code 26 "All other offenses" [that are less severe than the 25 higher order categories]) to check the robustness of our results. These results are presented in Tables A2 and A3. We find that, after removing each of these groups of crimes from our controls, our results remain fully consistent.

Given the differences we observe across cities, we next turn to non-deregulated cannabis offenses to more fully understand how decriminalization changed the cannabis offense landscape. Cannabis offenses that were not decriminalized in our cities fall into two categories: large quantity possession of cannabis and sale of cannabis. Evidence from cannabis legalization suggests that the impact of legalization on consumer demand is mixed, but unlikely to be negative. Small sample evidence on 24- to 27-year-olds suggests that use increases by 20% after recreational cannabis is legalized, while survey evidence on drug use suggests that legalization did not change usage among 18-year-olds (Zellers et al., 2022; Midgette and Reuter, 2020). Hollingsworth et al. (2022) also find that legalization increases use.

Results are presented in Tables 3 and 4. Beginning with Table 3, we observe significant changes in large quantity possession arrests, in particular with a large decrease of 23% in Los Angeles that is fully driven by decreases for arrests for Black and Hispanic individuals. Meanwhile, in New York, large quantity possession arrests increase, and they do so similarly for all three racial and ethnic groups. Though we cannot observe to what extent individuals or police officers changed their behavior after decriminalization, it is possible that the decreases we observe in Los Angeles and the increases we observe in New York could either result from police officers choosing to or being able to pursue fewer (or more) cannabis possession arrests more broadly, or from individuals changing their behaviors, such as those who were previously carrying large quantities of cannabis shifting to carrying amounts under the decriminalized limit or those who carried no cannabis before or carried it rarely starting to carry large quantities. The latter explanation would potentially be in line with the increasing usage rates documented post-legalization (Hollingsworth et al., 2022). The former, officers deprioritizing or being less able to detect illegal possession is evidenced by existing literature with regards to

Table 3
Cannabis decriminalization's effect on large quantity possession arrests.

	New York	Los Angeles	Chicago
Panel A: All Arrestees			
	50.68**	−70.43***	−7.72*
	(21.27)	(7.14)	(3.93)
Pre-Treatment Mean	70.00	305.25	47.62
Percent Change	72.41	−23.07	−16.20
F-test	0.61	0.75	0.00
N	592	600	752
Panel B: Black, Hispanic, and White Arrestees			
Black Arrestees	50.06**	−13.18*	−5.84
	(21.27)	(7.14)	(3.93)
Pre-Treatment Mean	37.12	97.75	37.12
Percent Change	134.84	−13.49	−15.74
F-test	0.25	0.12	0.01
N	592	600	752
Hispanic Arrestees	51.18**	−16.18**	1.16
	(21.27)	(7.14)	(3.93)
Pre-Treatment Mean	24.88	141.00	8.75
Percent Change	205.76	−11.48	13.23
F-test	0.16	0.18	0.00
N	592	600	752
White Arrestees	52.06**	5.82	5.78
	(21.27)	(7.14)	(3.93)
Pre-Treatment Mean	3.12	53.25	1.12
Percent Change	1665.89	10.92	514.01
F-test	0.07	0.05	0.10
N	592	600	752

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

legalization (Iannacchione and Ward, 2022; Henning et al., 2023), while future research would be valuable regarding the potentially changing choices individuals might make about the amount of cannabis they possess at any one time.

Looking at Table 4, we find that overall arrests for the sale of cannabis marginally increased in New York, and suggestively increased in Chicago. Considering the effects separated by racial and ethnic groups, the increase in New York is driven by increases for all racial and ethnic groups. In Los Angeles, sale arrests remain (suggestively) unchanged for all arrestees, as well as for Black individuals specifically.

In interpreting and contextualizing our results, one possible component as to why we may not see a decline in decriminalized possession arrests in Los Angeles, while we do in New York City and Chicago, is that decriminalization in Los Angeles took place at least four years before our other two cities. As discussed in Section 1, public support for cannabis legalization did not reach 50% support in America until 2011, but reached a two-thirds majority in 2020. Thus, it is possible that Los Angeles's 2010 decriminalization was less effective than Chicago's 2016 and New York's 2019 decriminalization in part because of a lack of social and police acceptance of cannabis usage at the time of implementation. Finally, when it comes to the cities' police departments, in all three cities, police departments were being investigated or assessed for potential biased policing when it comes to race in arrests in the year of or immediately after decriminalization. Both the Los Angeles Police Department and the Chicago Police Department were reprimanded by the US Department of Justice for concerns related to "excessive force and race" in their policing, while, the Office of the Inspector General of the New York Police Department released a report outlining the complaints that the New York Police Department received from 2014 to 2019, of those complaints, 68% regarded discrimination in policing, and 67% of the individuals who filed complaints were Black (Rubin, 2010; Goudie et al., 2017; del Valle, 2019).

4.1. The legalization of cannabis in New York City

To expand on our results, next we ask the question how the subsequent legalization impacted arrests for different racial and ethnic groups, focusing on New York City, the largest of the three study locations. As of 2023, 48% of Americans reside in a jurisdiction with legalized recreational use cannabis possession; however, the exact nature of what is legal differs by jurisdiction (Schaeffer, 2023).[13] In New York City, cannabis possession of up to 3 ounces (85 grams) (however, not the public consumption of cannabis[14])

[13] See Fig. A1 for a map capturing the differing legal weight limits across the country. Overall, higher social acceptance of cannabis usage is correlated with earlier legalization, and states without current legalization have the lowest social acceptance (Spetz et al., 2019).

[14] After cannabis legalization, adults over 21 are allowed to consume cannabis wherever smoking tobacco is allowed according to the smoke-free air laws (NYC Health, n.d.). The smoke-free air laws prohibit smoking any substance, including cannabis, in most public spaces (NYC Health, n.d.), therefore, we do not consider

Table 4
Cannabis decriminalization's effect on cannabis sale arrests.

	New York	Los Angeles	Chicago
Panel A: All Arrestees			
	41.68*	10.07	9.41**
	(21.27)	(7.14)	(3.93)
Pre-Treatment Mean	106.00	131.00	47.25
Percent Change	39.32	7.68	19.91
F-test	0.61	0.06	0.06
N	592	600	752
Panel B: Black, Hispanic, and White Arrestees			
Black Arrestees	43.43**	5.07	8.53**
	(21.27)	(7.14)	(3.93)
Pre-Treatment Mean	67.75	74.00	43.62
Percent Change	64.11	6.85	19.56
F-test	0.45	0.28	0.05
N	592	600	752
Hispanic Arrestees	50.18**	20.69***	5.66
	(21.27)	(7.14)	(3.93)
Pre-Treatment Mean	32.12	35.00	2.12
Percent Change	156.21	59.12	266.24
F-test	0.13	0.00	0.04
N	592	600	752
White Arrestees	51.43**	29.19***	4.28
	(21.27)	(7.14)	(3.93)
Pre-Treatment Mean	2.38	15.75	1.12
Percent Change	2165.64	185.34	380.68
F-test	0.07	0.00	0.04
N	592	600	752

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

became legal at the end of March 2021, when the bill was ratified and went into effect on the same day. As with decriminalization, the legalization of a specific action is relevant in cases well beyond drug regulation. Americans' attitudes and the legal landscape have shifted dramatically on a range of social issues over the 20th century. Therefore, understanding how effective legalization laws are in reducing the number of arrests is highly informative.

New York policymakers further empathized the goal of improving racial disparities as legislation moved beyond decriminalization and encompassed legalization. Then Gov. Andrew Cuomo said the day the law legalizing small quantity cannabis possession passed in New York that "the prohibition of cannabis disproportionately targeted communities of color [and] this landmark legislation provides justice for long-marginalized communities" (Mendez, 2021). Post-legalization, New York City Mayor Eric Adams framed the law as "a major step forward for equity and justice" (City of New York, 2023).

Table 5 summarizes the results for legalization in New York City. In Panel A, we find that legalization significantly decreases arrests across all arrestees as well as for each racial and ethnic group, suggesting that legalization effectively removed arrests as well as eliminated racial disparities for legalized offenses specifically. To complement the results here, and to capture more meaningful effect sizes in the presence of low pre-treatment observation counts, lastly, we re-estimate the model using a logistic regression, after converting each arrest category's monthly count outcome into a dummy. This substantially reduces the number of control groups with varying values, but enables us to better understand patterns in an outcome that largely takes up zero post-policy commencement. Looking at Panel B, we still observe that all arrests declined—though the effects for white arrestees is no longer significant due to the very low pre-policy frequency and large standard errors. We convert the reported log odds into percent changes, finding that the likelihood of arrest for small quantity cannabis possession decreased 94%–100% post-policy commencement.

Comparing legalization in New York City to decriminalization in New York City, we see that, while both policy levers decreased overall arrests, legalization brought them down to effectively zero while decriminalization led to a 39% decline. Looking at the effects of different legislative steps on each racial and ethnic group, we find that legalization in New York City decreased the arrests for all racial groups. These results suggest that legalization likely decreases overall arrests and racial disparities more than decriminalization does.

public consumption of cannabis as an offense category to be affected by legalization. (In practice, we observe the last public consumption charge-based arrest in the data in the month that decriminalization went into effect in 2019, therefore there are zero such charges in both the pre- and the post-period of the legalization analysis.) In addition, there is one offense category, defined as possession of 2–8 ounces of cannabis, that contains one ounce of weight range that became legal and five ounces of weight range that remained illegal. Given that distribution, we consider this offense category as untreated in the analysis.

Table 5
Cannabis legalization's effect on arrests for deregulated offenses in New York.

	All Arrestees	Black Arrestees	Hispanic Arrestees	White Arrestees
Panel A: OLS Regression				
	−55.26***	−53.64***	−52.39***	−51.76***
	(15.81)	(15.81)	(15.81)	(15.81)
Pre-Treatment Mean	4.62	2.75	1.25	0.38
F-test	0.38	0.23	0.47	0.29
N	592	592	592	592
Panel B: Logistic Regression				
	−3.45***	−3.45***	−2.78**	−38.94
	(1.33)	(1.33)	(1.26)	(142950169.95)
Percent Change	−96.83	−96.83	−93.83	−100.00
N	64	64	64	64

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.
Notes: Table 5 is constructed identically to Table 2, but with the treated group consisting of all cannabis offenses affected by legalization and treatment commencing at the time of legalization.

5. Conclusion

Cannabis arrests continue to be a quickly changing phenomenon in the United States, with high state-level variation in the legality of cannabis possession. Assessing the impact of these cannabis deregulation strategies is crucial to inform both the understanding of the current cannabis arrest landscape and the potential impact of future public policies. In this study, we use data from the three largest cities in the United States (New York City, Los Angeles, and Chicago), which all enacted policies decriminalizing cannabis possession. In addition to examining the effect of decriminalization in all three cities, we also consider the effect of cannabis use legalization in New York City.

We use a differences-in-differences approach to evaluate the impact of cannabis decriminalization. We find that decriminalization resulted in decreases in cannabis arrests overall in all three cities, though these reductions were not always driven by reductions in the specifically decriminalized possession weight categories. Commonly cited as a key, recent motivator for cannabis legalization is the policy goal of decreasing racial disparities in arrests, as Black and Hispanic individuals are more likely to be arrested for small quantity cannabis possession than their white counterparts. We observe that decriminalization narrowed racial disparities in arrests in Chicago by reducing small quantity possession arrests for Black individuals, and in Los Angeles by reducing large quantity possession arrests for both Black and Hispanic residents. We also observe that legalization decreased overall arrests for each racial group we consider, with similar impacts across groups.

We can consider the economic impact of these results on the one hand by assessing how these effects would translate to arrest reductions in a hypothetical federal possession decriminalization, and on the other by considering the costs of a possession conviction. While only a small share of all arrests involve cannabis possession, given the large number of such arrests nationwide in total—for example, 587,700 in 2016 and 317,793 in 2020 (Neill Harris, 2022; Ingraham, 2017)—a reduction could be substantial. Considering *all* possession arrests, we observe an approximate 13% reduction in such arrests after decriminalization of small quantity possession, which would translate to over 75,000 arrests averted per year compared to 2016, when many states have not decriminalized small quantity possession yet, or over 42,000 per year compared to 2020 when decriminalization was already increasingly widespread.[15]

On the other hand, we can consider the impact of a misdemeanor charge for those being convicted prior to decriminalization, which is sizable. Current estimates suggest that the annual earning loss associated with a misdemeanor conviction is $5100, which corresponds to a -16% earnings effect (Chien et al., 2022; Craigie et al., 2020). In comparison, studies generally find that for example the annual earnings penalty for motherhood is 5%–10% per child (Anderson et al., 2003; Kahn et al., 2014; Budig and England, 2001). Thus, a marijuana misdemeanor can have a considerable and persistent impact on the economic outcomes of a convicted individual. It is important to note that despite the reductions in racial disparities we documented as a result of decriminalization and legalization, because cannabis arrests make up a small share of all arrests, decriminalization can reduce but not eliminate racial disparities within the criminal justice system, which, evidence shows, exist across a range of facets of the criminal justice system, from traffic encounters through police use of force to judicial decisions and beyond (Goncalves and Mello, 2021; Lieberman, 2024; Arnold et al., 2025).

Policymakers' recent goals in deregulating cannabis possession were twofold—to decrease arrests for small quantity possession and to decrease racial disparities in the criminal justice system. Overall, our results suggest that the existing policies have made progress toward these goals. However, the varying impacts that decriminalization had on the number of arrests of Black and Hispanic individuals in our three cities suggest that these policies have not been as effective in reducing racial disparities in arrests as

[15] Because different states set different limits for what quantity of cannabis possession is decriminalized, and because we observe substantial changes for large quantity possession as well after decriminalization, in this back-of-the-envelope calculation we consider possession arrests with any drug weight, taking the average of the observed changes from Tables 2 and 3 where effects are significant and pre-trends are balanced.

policymakers had hoped, and there continue to be disparities in cannabis arrest rates across racial and ethnic groups. Meanwhile, legalization in New York City is able to effectively reach both policymaker goals of *fewer* and *fairer* arrests. Our results suggest that, while population-wide decriminalization of a certain crime may be partially effective as it decreases overall arrests, fully decreasing the racial inequities in the criminal justice system may require policies that take a more targeted approach.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

See Fig. A1 and Tables A1–A3.

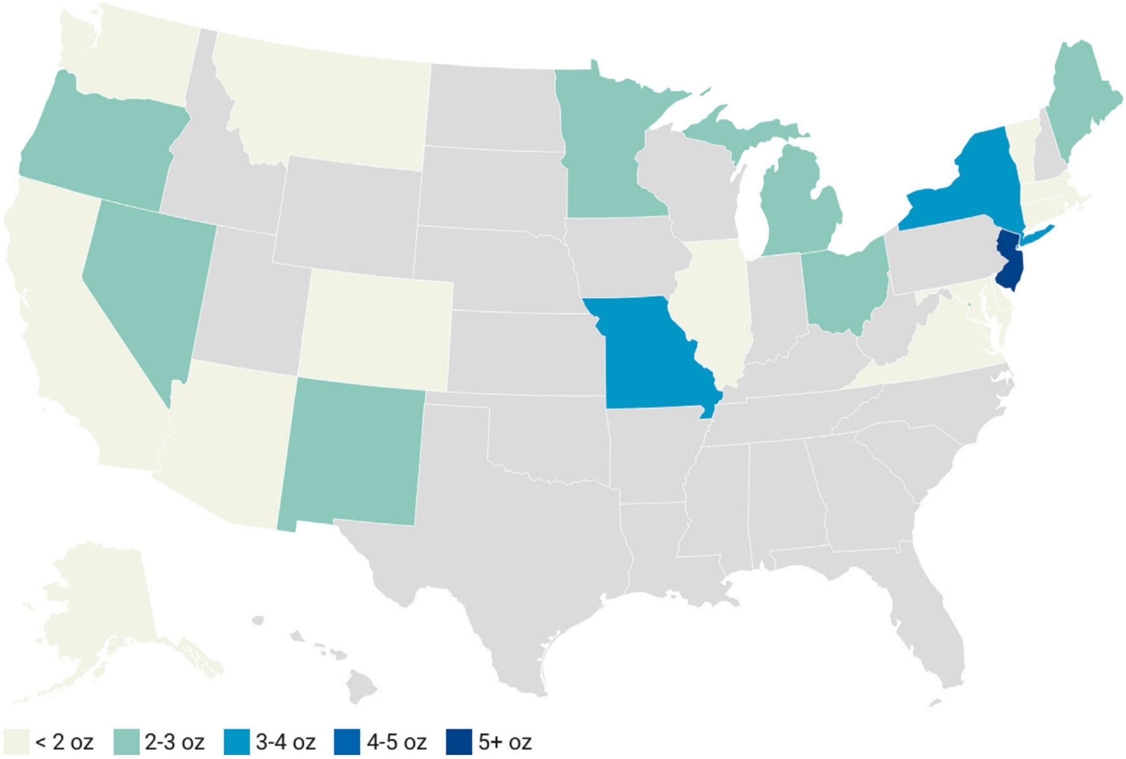


Fig. A1. Legal recreational cannabis possession limits by state.
Note: This figure is accurate as of July 29, 2025 (Williams, 2023; BeMiller, 2023; Jackson and Schaeffer, 2025).

Table A1
Decriminalized cannabis offense and the commencement of decriminalization.

	New York City	Los Angeles	Chicago
Panel A: Decriminalized Cannabis Charges			
Weight up to	57 g	28.5 g	10 g
Additional charge categories	Public Consumption		
Panel B: Ratification and Effectiveness Dates			
Ratification date	07/29/2019	9/30/2010	7/29/2016
Effectiveness date	08/28/2019	1/1/2011	7/29/2016

Notes: In New York City, decriminalization lowered the penalty for possession of cannabis of up to 57 g as well as public consumption from a misdemeanor to an infraction (New York Police Department, 2018; The Brooklyn District Attorney's Office, 2018; Mueller, 2018; Campbell, 2019). In Los Angeles and Chicago, decriminalization lowered penalties from a misdemeanor to a civil infraction (Condon, 2010; Illinois General Assembly, n.d.). In all cases, policies were ratified (i.e., when the state's governor signed the bill into law) either before or in conjunction with their effectiveness date (i.e., when the law took effect). The dates are determined for each city from the following sources: New York City: (Campbell, 2019); Los Angeles: (Condon, 2010; Office of the District Attorney, n.d.); Chicago: (Garcia, 2016b; Illinois General Assembly, n.d.).

Table A2
Cannabis decriminalization's effect on arrests for decriminalized offenses—robustness test excluding the lowest level offenses.

	New York	Los Angeles	Chicago
Panel A: All Arrestees			
	−53.84**	−5.95	−14.26***
	(21.89)	(6.82)	(3.74)
Pre-Treatment Mean	136.75	75.75	20.62
Percent Change	−39.37	−7.86	−69.12
F-test	0.59	0.01	0.81
N	576	588	736
Panel B: Black, Hispanic, and White Arrestees			
	(21.89)	(6.82)	(3.74)
Pre-Treatment Mean	83.50	21.50	17.00
Percent Change	−15.53	58.37	−64.74
F-test	0.68	0.03	0.56
N	576	588	736
Hispanic Arrestees	17.16	8.92	0.37
	(21.89)	(6.82)	(3.74)
Pre-Treatment Mean	43.00	34.00	2.88
Percent Change	39.91	26.25	12.85
F-test	0.31	0.81	0.04
N	576	588	736
White Arrestees	48.04**	11.67*	2.49
	(21.89)	(6.82)	(3.74)
Pre-Treatment Mean	5.88	17.25	0.62
Percent Change	817.63	67.68	399.11
F-test	0.11	0.00	0.04
N	576	588	736

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A3
Cannabis decriminalization's effect on arrests for Ddecriminalized offenses—robustness test excluding the highest level offenses.

	New York	Los Angeles	Chicago
Panel A: All Arrestees			
	−51.49**	−3.09	−12.59***
	(21.80)	(7.29)	(4.11)
Pre-Treatment Mean	136.75	75.75	20.62
Percent Change	−37.65	−4.07	−61.06
F-test	0.66	0.01	0.91
N	576	588	720
Panel B: Black, Hispanic, and White Arrestees			
Black Arrestees	−10.61	15.41**	−9.34**
	(21.80)	(7.29)	(4.11)
Pre-Treatment Mean	83.50	21.50	17.00
Percent Change	−12.71	71.69	−54.96
F-test	0.61	0.02	0.65
N	576	588	720
Hispanic Arrestees	19.51	11.79	2.03
	(21.80)	(7.29)	(4.11)
Pre-Treatment Mean	43.00	34.00	2.88
Percent Change	45.38	34.67	70.65
F-test	0.27	0.57	0.03
N	576	588	720
White Arrestees	50.39**	14.54*	4.16
	(21.80)	(7.29)	(4.11)
Pre-Treatment Mean	5.88	17.25	0.62
Percent Change	857.69	84.28	665.00
F-test	0.09	0.00	0.04
N	576	588	720

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Data availability

The authors do not have permission to share data.

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